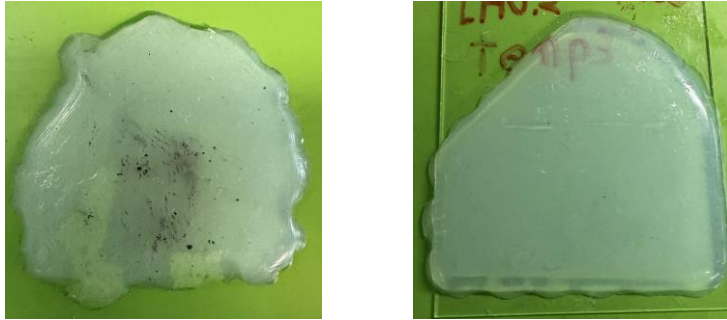


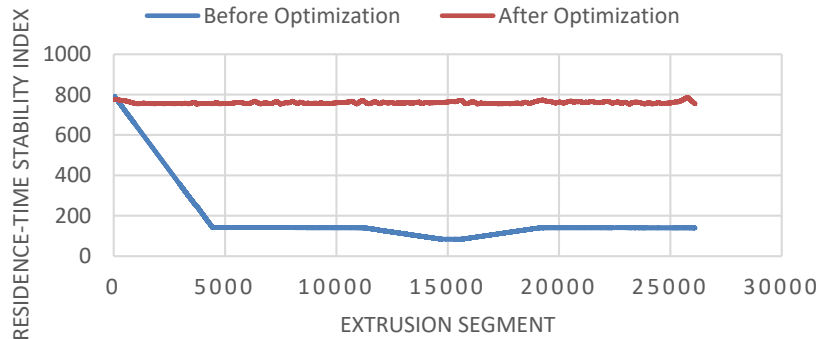
# Advanced Additive Manufacturing Research (Ongoing)

## Model-Based Optimization of Silicone Additive Manufacturing

### Printing Results Before vs After Optimization



### IMPROVED MATERIAL CONDITIONING STABILITY



### OVERVIEW

Research on process modeling for low-viscosity silicone extrusion in additive manufacturing. The work links residence time and thermal history to material evolution in order to improve flow consistency and print quality.

### MY CONTRIBUTIONS

- Developed residence-time tracking model
- Modeled dynamic material behaviour
- Developed predictive process modeling framework
- Connected material state to process performance
- Applied PID-based control for extrusion stability
- Evaluated optimization effects on print quality

### RESULT

- Improved material conditioning consistency
- Reduced extrusion instability during printing
- Validated key process factors affecting print quality
- Established framework for process optimization
- Ongoing journal paper preparation